ISONIAZID

Give with pyridoxine

Hepatotoxic

CHEMICAL NAME Pyridine-4-carbohydrazide
CLASS First-line antituberculosis drug (hydrazide)
TRADE NAMES Rimifon · Nydrazid · Isoniazid (generic) · Rifinah (with rifampicin)

THERAPEUTIC USES / INDICATIONS

- Tuberculosis treatment (all forms) in combination: 2 months HRZE then 4 months HR.
- TB preventive therapy (latent TB); alone or with rifapentine/rifampicin.

ADMINISTRATION

- **PO** on an empty stomach (1 h before or 2 h after food), once daily, usually in fixed-dose combination.
- **IM/IV** when oral not possible.
- Directly observed therapy (DOT) recommended.

DOSAGES

- **Adult:** 5 mg/kg once daily (max 300 mg).
- **Children:** 10 mg/kg (7–15) once daily (max 300 mg).
- **Latent TB:** 300 mg daily for 6–9 months.
- Give pyridoxine 10–25 mg daily to prevent neuropathy.

PHARMACOKINETICS

- **Peak:** 1–2 h. Good CSF penetration.
- Metabolised in liver by N-acetylation (slow vs fast acetylators — slow acetylators at higher toxicity risk).
- Excreted in urine. **t½:** 1–4 h depending on acetylator status.

PHARMACODYNAMICS

- Inhibits synthesis of **mycolic acids** in mycobacterial cell wall → bactericidal against actively dividing *M. tuberculosis*.
- Inhibits CYP2C19, CYP3A4.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, epigastric discomfort.
- Rash, acne.
- Mild transient rise in liver enzymes.

ADVERSE EFFECTS (SERIOUS)

- Hepatitis (risk ↑ with age, alcohol, rifampicin, pyrazinamide).
- Peripheral neuropathy (pyridoxine deficiency; ↑ in diabetes, alcohol, malnutrition, HIV, pregnancy).
- Seizures, psychosis, optic neuritis.
- Drug-induced lupus; agranulocytosis.

CONTRAINDICATIONS & CAUTIONS

- Previous isoniazid-induced hepatitis.
- Acute liver disease.
- **Caution:** chronic liver disease, alcohol dependence, epilepsy, HIV, slow acetylators.

KEY INTERACTIONS

- **Phenytoin, carbamazepine:** ↑ levels → toxicity.
- **Antacids:** ↓ absorption.
- **Paracetamol, alcohol:** ↑ hepatotoxicity.
- **Tyramine/histamine-rich foods (cheese, fish):** flushing, palpitations (rare).

NURSING MANAGEMENT

- Baseline LFTs; monitor monthly in high-risk patients.
- Ensure pyridoxine is co-prescribed.
- Ask about nausea, fatigue, dark urine, jaundice; stop and report.
- Check numbness or tingling in hands/feet.
- Support adherence (DOT); notify TB programme; contact tracing.

PATIENT EDUCATION

- Take every day on an empty stomach for the full course (usually 6 months).
- Avoid alcohol.
- Report loss of appetite, vomiting, yellow eyes, dark urine or tingling at once.
- Cover mouth when coughing; ventilate rooms.

KEY POINTS

- First-line TB drug (H); mycolic acid inhibitor.
- Hepatotoxicity — monitor LFTs.
- Always give pyridoxine (neuropathy).
- 5 mg/kg (max 300 mg) daily.